

NMAM INSTITUTE OF TECHNOLOGY, NITTE*(An Autonomous Institution affiliated to VTU, Belagavi)***Eighth Semester B.E. (CSE) (Credit System) Degree Examinations**

May - June 2023

19CSE75 / 17CS604 - BUSINESS INTELLIGENCE

Duration: 3 Hours

Max. Marks: 100

*Note: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II** each and **One full** question from **Unit – III**.*

Unit – I		Marks	BT*	CO*	PO*
1. a)	Write various sources and characteristics of structured, unstructured and semi-structured data.	10	L*1	1	1
b)	What is BI? Explain BI component framework with neat diagram.	10	L2	2	1
2. a)	What are OLTP and OLAP? Give detailed explanation for both.	10	L1	1	1
b)	What is data warehouse? Explain Ralph Kimball's and Inmon's approach	10	L2	2	1
3. a)	With Example explain the STAR and Snowflake data model for OLAP.	10	L2	1	1
b)	What is ETL? Explain intermediate stages of ETL with neat Diagram.	10	L2	2	1
Unit – II					
4. a)	What is data profiling? When and how to conduct data profiling? Explain.	10	L2	3	1
b)	Illustrate different types of data model.	10	L2	4	1
5. a)	What is dimension table? Explain any three types of dimension table.	10	L1	3	1
b)	What are dashboards? How do you create dashboards? Explain.	10	L2	4	1
6. a)	Write the features of good reporting.	10	L2	3	1
b)	Define KPI's. Identify and Illustrate salient features of good metric.	10	L3	4	1
Unit – III					
7. a)	What is data mining? With diagram explain the steps involved in the process of extracting information from the data.	10	L1	5	1
b)	What is frequent itemset mining? Explain with market basket analysis in detail.	10	L2	5	1
8. a)	Explain following terms:				
i.	Classification				
ii.	Regression				
iii.	Clustering analysis				
iv.	outlier analysis	10	L2	5	1
b)	Explain apriori algorithm for finding frequent patterns by confined candidate generation.	10	L2	5	2

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
